Rajat Sawanni, Ph.D.

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Research Interests

Sustainable Aviation and Emissions
Soot formation mechanisms, contrail formation, rocket emissions, non-volatile and volatile particulate emissions

Combustion Systems and Technologies
Sustainable Aviation Fuels, Hydrogen, Laminar combustion systems including high pressure coflow, and counterflow diffusion flames, turbulent combustion systems including spray combustion, rocket engines, and model gas turbine combustors

Experimental Diagnostics
Particle imaging velocimetry (PIV), Rayleigh and Mie scattering, shadowgraphy, schlieren imaging, extinction and emission spectroscopy, Multi-Angle Light Scattering (MALS), Laser Induced Fluorescence (LIF), Laser Induced Incandescence (LII). thermophoretic sampling, aerodynamic sampling, electron microscopy, gas chromatography, mass spectroscopy

Analytical Methods
Abel Inversion techniques, Radiative transfer analysis, Monte Carlo methods, Network theory, Kinetic model reduction, Numerical modeling

Education

Doctor of Philosophy (Ph.D.) in Aerospace Engineering May 2025
University of Toronto, Ontario, Canada
Institute for Aerospace Studies
Thesis: A combined experimental and numerical study of pressure dependence of sooting processes in a counterflow diffusion flame

Bachelor and Master of Technology in Mechanical Engineering Aug 2018
Indian Institute of Technology, Bombay, Mumbai, India
Specialization in Thermal and Fluids Engineering (TFE)
Thesis: Injector design and characterizations for Gas-Liquid Rocket Engine

Research Experience

Post Doctoral Fellow Jun 2025 – current
Institute for Aerospace Studies, University of Toronto
Contributing to three interdisciplinary projects spanning multiple areas of sustainable aviation technologies, working collaboratively with academic researchers and industry partners. Support includes development and operation of experimental facilities, academic writing, research coordination and supervision, and developing funding proposals.

- **Contrails-in-a-lab: Unraveling the multi-physical pathways of contrail inception and growth**
  Principal Investigators: Prof. Ömer L. Gülder, Prof. C. P. T. Groth, Prof. S. Chaudhuri, in collaboration with Pratt & Whitney Canada and NSERC Canada.

  *Generating high-quality research data on contrails through optical diagnostics of multi-phase flows utilizing techniques such as particle image velocimetry (PIV), Rayleigh scattering, multi-angle light scattering, absorption, and extinction for unraveling the mechanism of contrail formation. The goal is to conduct tractable experiments with state-of-the-art non-intrusive diagnostics, in order to guide future aviation technologies on **climate change mitigation** and improve the understanding and modeling of these multi-physical processes.*

- **Emission characterization of Sustainable Aviation Fuels in liquid bi-propellant rocket engines**
  Principal Investigators: Prof. Ömer L. Gülder, in collaboration with NordSpace

  *Contributing to advance Canada's first **carbon-neutral commercial launch vehicle** by experimentally evaluating Sustainable Aviation Fuels (SAF) in the Hadfield liquid bi-propellant rocket engine (NordSpace). The work involves characterizing engine performance and emissions during SAF/LO_x hot-fire tests and informing injector and cooling-loop redesigns to improve SAF compatibility, with the overarching goal of enabling flexible, low-carbon **green engines** for future launch vehicles.*

- **Impact of hydrogen addition on the sooting tendency of ethylene counterflow diffusion flames**
  Principal Investigators: Prof. Ömer L. Gülder, in collaboration with Prof. Emre Karataş

  Isolating and investigating the chemical effects of adding hydrogen to ethylene counterflow diffusion flames at high pressure. The work aims to extend our understanding of the role of hydrogen in the sooting tendencies of hydrocarbon fuels at high pressure.

Ph.D. Candidate Sep 2018 – May 2025
Institute for Aerospace Studies, University of Toronto
Contributed to the design and development of research facilities focusing on high pressure counterflow flames and low pressure, low temperature contrails. Work included the design, development and operation of experimental facilities, numerical and analytical tools, academic writing, research conceptualization and coordination.

- **A Combined Experimental and Numerical Study of Pressure Dependence of Sooting Processes in a Counterflow Diffusion Flame**
  Advisor: Prof. Ömer L. Gülder

  Investigated the pressure dependence of sooting processes by designing experimental, numerical and analytical tools for counterflow diffusion flames. The goal was to unify the disparate pressure dependence of sooting processes in ethylene soot forming counterflow diffusion flames across various dilutions, strain rates and carbon flux.

- **Contrails-in-a-lab: Design and development of a novel laboratory scale facility for generating and analyzing contrail cirrus**
  Principal Investigators: Prof. Ömer L. Gülder, Prof. C. P. T. Groth, Prof. S. Chaudhuri, in collaboration with Pratt & Whitney Canada and NSERC Canada.

  *The design and development of a **novel experimental facility** that generates laboratory ice condensation trails (contrails) by reproducing heterogeneous ice nucleation arising from the turbulent mixing of a high-temperature combustion exhaust jet (*~ 600 K*) with a low-pressure (*~ 0.2 bar*), cold ambient flow (*~ 210 K*).*

Master's Candidate Jan 2015 – April 2018
Department of Mechanical Engineering, IIT Bombay
Worked on various independent research projects as a bachelor's and masters student, gaining expertise in various fields of thermo-fluid sciences.

- **Development of Gas–Liquid Rocket Engines** May 2016 – Apr 2018
  Advisor: Prof. Arindrajit Chowdhury, IIT Bombay

  Contributed to the development and operation of a gas–liquid rocket engine test facility, including the design, fabrication, and characterization of multiple single-element injector configurations to evaluate novel fuels and advance engine technologies.

- **Flow Induced Reconfiguration of Aquatic Vegetation** Jan 2015 – Dec 2016
  Advisor: Prof. Rajneesh Bhardwaj, IIT Bombay

  Conducted numerical research on the deformation of flexible cilia using an in-house Fluid–Structure Interaction solver, extending this work into a theoretical framework that advances the understanding of flow-induced reconfiguration in aquatic vegetation under low Reynolds number viscous conditions.

Teaching and Research Pedagogy

Course Instructor, Combustion Processes Sep 2020 – Dec 2020
AER315, University of Toronto

Designed and delivered the core undergraduate course Combustion Processes for third-year Engineering Science students, including curriculum and course material development, assessment design, and student support in a fully remote teaching environment.

Teaching Assistant Jul 2017 – Dec 2023
University of Toronto & Indian Institute of Technology Bombay

Supported undergraduate instruction across mathematics, thermofluids, and combustion courses through tutorial delivery, assessment design and grading, course content management, and student support. Courses and their details are listed below

- **Calculus I (ESC194)** – University of Toronto
- **Calculus II (ESC195)** – University of Toronto
- **Fundamentals of Combustion (AER1304)** – University of Toronto
- **Thermodynamics (ME209)** – IIT Bombay
- **Mechanical Measurements (ME226)** – IIT Bombay

Research Supervision and Mentoring 2025 – current
University of Toronto

Co-supervised and Mentored undergraduate and graduate students.

- Anay Shah - Undergraduate researcher from University of Toronto, now pursuing Masters at Imperial College, London, UK. Final year thesis: Dependencies of light scattering properties from soot and ice particles in contrail cirrus.
- Arya Ollivierra - Undergraduate Intern from Princeton University. Project: Design and development of a sampling system for in-situ and ex-situ analysis of soot and ice particles in the contrail facility.
- Hannah Drouillard - Undergraduate researcher from University of Toronto. Project: Direct numerical simulation of scalar mixing leading to ice nucleation in contrail cirrus.
- Aryan Nobakht - Doctoral candidate from Toronto Metropolitan University. Project: Impact of hydrogen addition to sooting tendencies of high pressure counterflow diffusion flames.
- Junhee Park - Undergraduate researcher, University of Toronto. Project: Large Eddy Simulations of supersaturation fluctuations through jet mixing in contrail relevant conditions.
- George Chen - Undergraduate researcher, University of Toronto. Project: Development of a wavelength modulated tunable diode laser absorption spectroscopy (WM-TDLAS) system for water vapor measurement in the contrail tunnel.
- Necdet Tektunali - Masters researcher, University of Toronto. Project: Volatile particulate matter (vPM) and non-volatile particulate matter (nvPM) measurements of combustion exhausts.

Academic services

- Manuscript peer review - **Combustion and Flame**
- Member - Combustion Institute

JOURNAL ARTICLES

1. **Sawanni, R.***, Roy, A.*, Rajan, Y., Mayigue, C. C., Taddesse, T., Jahncke, I., Chaudhuri, S., Groth, C. P. T., Gülder, Ö. L. (2026). A contrail-in-a-lab facility to study the inception and development of contrails. *Bulletin of American Meteorological Society.* (in preparation)
2. Roy, A.*, **Sawanni, R.***, Rajan, Y., Mayigue, C. C., Taddesse, T., Jahncke, I., Chaudhuri, S., Groth, C. P. T., Gülder, Ö. L. Effects of fuel and soot characteristics on the inception and development of contrails. *Proceedings of the Combustion Institute.* (accepted)
3. **Sawanni, R.**, and Gülder, Ö. L. Effects of pressure on the chemical sooting structure of equi-diffusive counterflow diffusion flame. *Proceedings of the Combustion Institute.* (accepted)
4. **Sawanni, R.**, and Gülder, Ö. L. (2026). Influence of pressure on the structure and chemistry of soot formation counterflow diffusion flame. *Combustion and Flame.* 286, 114873.
5. **Sawanni, R.**, and Gülder, Ö. L. (2025). Retrieval of temperature and extinction coefficient from modulated absorption emission technique: Effects of practical light collection, self-absorption, in-scattering, and finite spatial resolution. *Journal of Quantitative Spectroscopy and Radiative Transfer.* 347, 109662.
6. **Sawanni, R.**, and Gülder, Ö. L. (2024). A tractable methodology for assessing the pressure scaling of sooting processes in a counterflow diffusion flame from 1 to 6 bar. *Proceedings of the Combustion Institute*, 40(1-4), 105745.
7. Bhati, A.*, **Sawanni, R.***, Kulkarni, K., and Bhardwaj, R. (2018). Role of skin friction drag during flow-induced reconfiguration of a flexible thin plate. *Journal of Fluids and Structures*, 77, 134–150.

REFEREED CONFERENCE PAPERS

1. Jahncke, I., Mayigue, C. C., Taddesse, T., Groth, C. P. T., Roy, A., **Sawanni, R.**, Rajan, Y., Chaudhuri, S., Gülder, Ö. L. (2026). FANS-based predictions of jet-regime contrail formation using a two-equation soot/ice aerosol transport model. *AIAA SciTech 2026 Forum.* (submitted)
2. Mayigue, C. C., Taddesse, T., Jahncke, I., Groth, C. P. T., Roy, A., **Sawanni, R.**, Gülder, Ö. L., Chaudhuri, S. and Rajan, Y. (2025). Contrail formation simulation via FANS-based turbulence modeling combined with two-equation soot/ice particle transport modeling. *AIAA SciTech 2025 Forum*, 2025-0600.

OTHER CONFERENCE PAPERS AND TALKS

1. Roy, A.*, **Sawanni, R.***, Rajan, Y., Mayigue, C. C., Taddesse, T., Jahncke, I., Chaudhuri, S., Groth, C. P. T., Gülder, Ö. L. Effects of fuel and soot characteristics on the inception and development of contrails. *International Symposium on Combustion*, Kyoto, Japan, 2026
2. **Sawanni, R.**, and Gülder, Ö. L. Effects of pressure on the chemical sooting structure of equi-diffusive counterflow diffusion flame. *International Symposium on Combustion*, Kyoto, Japan, 2026
3. **Sawanni, R.**, and Gülder, Ö. L. (2025). Investigation of chemical sooting structures in counterflow diffusion flames from 1 to 6 bar. *Combustion Institute's Canadian Section Spring Technical Meeting*, University of Calgary, Calgary, Canada
4. Jahncke, I., Mayigue, C. C., Taddesse, T., Groth, C. P. T., Roy, A., **Sawanni, R.**, Rajan, Y., Chaudhuri, S., Gülder, Ö. L. (2025). FANS-based predictions of contrail formation

*Contributed equally

in the jet-regime using a two-equation soot/ice aerosol transport model. *Combustion Institute's Canadian Section Spring Technical Meeting*, University of Calgary, Calgary, Canada

5. **Sawanni, R.**, and Gülder, Ö. L. (2024). A tractable methodology for assessing the pressure scaling of sooting processes in a counterflow diffusion flame from 1 to 6 bar. *International Symposium on Combustion*, Milan, Italy, 2024
6. **Sawanni, R.**, and Gülder, Ö. L. (2024). A tractable methodology for assessing the pressure scaling of sooting processes in a counterflow diffusion flame from 1 to 6 bar. *Combustion Institute's Canadian Section Spring Technical Meeting*, Queen's University, Kingston, Canada
7. **Sawanni, R.**, and Gülder, Ö. L. (2024). Pressure Scaling of Sooting Processes in a Counterflow Diffusion Flame from 1 to 6 bar. *Twenty-first International Conference on Flow Dynamics*, Institute for Fluid Sciences Tohuku University, Sendai, Japan
8. **Sawanni, R.**, and Gülder, Ö. L. (2023). Assessment of tractability of laminar opposed-flow diffusion flames for studying pressure dependence of sooting processes. *Combustion Institute's Canadian Section Spring Technical Meeting*, University of Alberta, Alberta, Canada
9. **Sawanni, R.**, and Gülder, Ö. L. (2022). Optical considerations for low soot measurements in steady state counterflow diffusion flames using Diffuse Back Illuminated Extinction Imaging - Temperature Imaging at high pressures. *Combustion Institute's Canadian Section Spring Technical Meeting*, University of Ottawa, Ottawa, Canada
10. **Sawanni, R.**, and Gülder, Ö. L. (2022). Tractability of laminar counterflow diffusion flames to assess the pressure dependence of sooting processes. *Combustion Institute's Canadian Section Spring Technical Meeting*, University of Ottawa, Ottawa, Canada
11. Shaikh, T., **Sawanni, R.**, Chowdhury, A., Umakant, S., (2017). A comparison of the performance of uni-element injectors for laboratory scale gas-liquid rocket engines, 11^{th} *Asia Pacific Conference on Combustion*, Sydney, Australia

Technical Skills

Combustion system design	Coflow and counterflow high pressure burners, Model gas turbine combustors, liquid rocket engines, turbulent premixed burners
Ex-situ techniques	thermophoretic sampling, aerodynamic sampling, exhaust gas analysis, mobility sizing, phase doppler anemometry, electron microscopy, gas chromatography, mass spectroscopy, optical particle counters
Optical diagnostics	Non-intrusive optical diagnostics like particle imaging velocimetry (PIV), Rayleigh and Mie scattering, shadowgraphy, schlieren imaging, extinction and emission spectroscopy, multi-angle light scattering, laser induced fluorescence, laser induced incandescence
Process control	PID integration, flow control and automation, cryogenic system design, high pressure and vacuum system design, thermal management systems
Analytical Languages	Python, MATLAB
Design, modeling and analysis	Auto CAD, SolidWorks, Inventor, GMSH, MATLAB/Simulink, OpenFOAM, Tecplot, ANSYS, LABVIEW Signal Expres, Mathematica, CHEMKIN, Cantera, OpenSMOKE++
Analytical modeling	Fluid structure interactions, Abel inversion, Radiative transfer analysis, Monte Carlo methods, global reaction pathway analysis

Scholastic Achievements

Proficiency Tests

- IELTS General Training: **8.5/9.0** 2025
- Graduate Record Examination (GRE): **327/340** 2018

Awards and Accolades

- Best presentation Award for Young Researcher - 21st International Conference on Flow Dynamics, Tohuku, Japan 2024
- Undergraduate Research Award, Indian Institute of Technology Bombay 2017
- National Physics and Astronomy Olympiads (NSEP & NSEA): Ranked in the national top 1% (among more than 200,000 students), India 2013

Community Service

- Volunteered for one year with the National Service Scheme (NSS)
- Raised awareness and funds for the National Association for the Blind, India
- Organized a campus-wide donation drive for neighboring underprivileged communities
- Promoted sustainable energy technologies in rural India as a volunteer with the Yusuf Meherally Centre, supporting small-scale industrial development